Fiscal Year:	FY 2018	**Task Last Updated:**	FY 07/25/2017
PI Name:	Hofmann, Douglas Ph.D.		
Project Title:	Study of Mushy-Zone Development in Dendritic Microstructures with Glass-Forming Eutectic Matrices		
Division Name:	Physical Sciences		
Program/Discipline:	MATERIALS SCIENCE		
Program/Discipline--Element/Subdiscipline:	MATERIALS SCIENCE--Materials science		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight,Ground	**Solicitation / Funding Source:**	2010 Materials Science NNH10ZTT001N
Start Date:	10/01/2010	**End Date:**	09/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	1	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA MSFC
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Flight Program:			
Flight Assignment:	NOTE: Extended to 9/30/2019 per PI (Ed., 7/5/16) NOTE: Extended to 10/1/16 (Ed., 7/31/13) NOTE: To be extended past 10/1/2013 per PI (Ed., 7/24/13)		
Key Personnel Changes/Previous PI:	August 2017: None.		
COI Name (Institution):	Johnson, William Ph.D. (California Institute of Technology) Shapiro, Andrew Ph.D. (Jet Propulsion Laboratory) Roberts, Scott Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	Internal Project		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The future materials for NASA spaceflight applications are likely to be high-performance composites. Metallic glass matrix composites, which are metallic glasses reinforced with crystalline dendritic inclusions, were found to impart benchmark properties (see for example, D.C. Hofmann et al. Nature, 451:1085, 2008). The mechanical performance of these composites is understood to be highly dependent on the morphological-feature length scales of the dendritic microstructure. Specifically, it has been demonstrated that the length scales associated with the size and distribution of dendrites has a profound effect on mechanical properties such as ductility, fracture toughness, and fatigue endurance. Understanding and controlling the development of the dendrite microstructure will enable optimization of the mechanical properties and promote a truly unique mechanical performance of these composites. The investigators propose to study the nucleation, growth, directionality, and cooling rate dependence of crystalline dendrites evolving in a glass-forming eutectic matrix. The ground-based research effort will be carried out utilizing laboratories at the Jet Propulsion Laboratory and the California Institute of Technology, including the High Vacuum ElectroStatic Levitator platform (HVESL) at Caltech. The investigators propose a four-component ground-based project consisting of (1) directional solidification experiments utilizing Moving Mushy Zone (MMZ) and Bridgman solidification, (2) morphological characterization of dendrite growth and orientation during "mushy" (or "semi-solid") processing, (3) mechanical testing and development of structure-property relations, and (4) dendrite formation and growth during containerless processing utilizing the HVESL. After completion of a ground-based research program, the investigators will extend the experimentation to flight-based microgravity research aboard the International Space Station (ISS), the aim of which is to study gravity-driven convection arising from density gradients in the microstructure. The flight experiment objectives are to (1) use the Low Gradient Furnace (LGF) to complete Bridgman crystal growth on composites in an attempt to grow dendrites along the translational axis, (2) use the Solidification and Quenching Furnace to coarsen and homogenize dendrite microstructures using mushy-zone processing.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The development of innovative new materials is a fundamental part of future terrestrial and space technologies. These materials have a tremendous benefit towards improving life on Earth, from improving the accessibility and cost of solar power, to making cars safer and stronger, to reducing the mass of aircraft, for example. Innovative technologies typically require new materials tailored for those applications. This drives materials development. Metal matrix composites (MMCs) are a relatively new class of metal alloys that exhibit unique properties which make them desirable for many new technologies. MMCs are typically two-phase alloys consisting of a continuous metal matrix incorporated with a second phase (which can be a metal dendrite, a fiber, a particle, or some other inclusion). They differ from traditional monolithic crystalline metal alloys because the mechanical and physical properties of single-phase alloys are limited by composition and dislocation-based plasticity. By creating a composite phase, MMCs have the ability to combine the beneficial properties of both phases into a single alloy, like strength and toughness, for example. A novel type of MMC has been recently developed that exhibits benchmark combinations of strength and toughness. These alloys were developed by creating a MMC with high-strength metallic glasses (also called amorphous metals) as the matrix phase and a soft body centered cubic (bcc) dendrite as the crystalline phase. These metallic glass matrix composites (MGMCs), which are the focus of the current research effort, have been shown to have the strength of carbides, with the toughness and fatigue limit of stainless steel and the density of titanium. These alloys have the potential for significant impact in high-performance structural applications in areas ranging from defense, to aerospace, to commercial sporting goods. The widespread use of MMC (and MGMCs in particular) requires in-depth research into the processing and fabrication of these multi-phase alloys. For example, semi-solid (or mushy) processing is often required to coarsen the second phases in MMCs and MGMCs to improve the mechanical properties. These additional processing requirements make the fabrication of hardware from MMCs unique compared with traditional metal parts (which are typically machined into a final shape). The NASA-funded technology being developed as part of the current program seeks to understand the development of second phases in MGMCs as a function of processing parameters so that innovative industrial processing techniques can be used to generate hardware. The end result is the potential widespread use of MMC in applications such as biomedical implants or spacecraft shielding.
Task Progress:	The focus of this project is to use the capabilities of the ISS to perform semi-solid holds at 1,200°C on a variety of bulk metallic glass matrix composites (BMGMCs) reinforced with different volume fraction, morphology, and composition of particles with vastly different densities than the bulk metallic glasses (BMG) matrix. The objective is to study the wear resistance as measured through pin-on-disk testing, the morphology of the crystalline phases processed in microgravity, and the toughness as measured through indentation-initiated crack growth. The expectation is that by reducing sedimentation in the mushy zone, a bimodal and homogeneous distribution of dendrites can be obtained that imparts improved wear resistance and crack resistance compared with samples of the same composition (but different microstructure) being produced in 1-g. This will be achieved by reinforcing the inter-particle region of the composite with dendrites, while still having a dilute volume fraction of particles (<50% by volume). A comparison will be made between a BMG that becomes brittle when crystalline (Vitreloy 106) and one that becomes soft when crystallize (CuZrAlCo). The experiment will be used to identify optimal microstructures for improving the properties of crystalline BMGMCs. The ISS facility is ideal for the current experiment because it can provide a stable processing temperature of 1,200°C (1,473 K) for long periods of time in an oxygen free environment with low-gravitational effects. The relatively slow quenching rate on the ISS furnace is not sufficient to form fully amorphous samples (centerline cooling must be greater than 1,000 K/s in the sample) but the current experiment relates to forming crystalline composites and therefore the ISS is a suitable facility for what is proposed. In the current year of funding, ground truth testing was used to validate the hypothesis of the flight experiment and a science concept review for the flight experiment was prepared.
Bibliography Type:	Description: (Last Updated: 01/01/2026)
Articles in Peer-reviewed Journals	Hofmann DC, Polit-Casillas R, Roberts SN, Borgonia JP, Dillon RP, Hilgemann E, Kolodziejska J, Montemayor L, Suh JO, Hoff A, Carpenter K. "Castable bulk metallic glass strain wave gears: towards decreasing the cost of high-performance robotics." Sci Rep. 2016 Nov 24;6(1):37773. https://doi.org/10.1038/srep37773 , Nov-2016